# CS 4391 Introduction to Computer Vision
# Homework 1

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Download the homework1_programming.zip file from eLearning, Assignments, Homework 1. Finish the following programming problems and submit your scripts to eLearning. You can zip all the data and files for submission. TA will run your scripts to verify them.

Install the Python packages needed by

- pip install -r requirement.txt

Here are some useful resources:

- Python basics `https://pythonbasics.org/`
- Numpy `https://numpy.org/doc/stable/user/basics.html`
- OpenCV `https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html`

## Problem 1

(5 points) Image gradient using central difference.

Compelet the main() function in image_gradient.py.

This script first reads the "cracker_box.jpg" image, and then performs central difference to compute the gradient of the image. The equations of the central difference are

$$\frac{\partial I(x,y)}{\partial x} = \frac{I(x+1,y) - I(x-1,y)}{2}, \tag{1.1}$$

$$\frac{\partial I(x,y)}{\partial y} = \frac{I(x,y+1) - I(x,y-1)}{2}, \tag{1.2}$$

where $I(x,y)$ denotes the image. Note: in this implementation, you cannot directly use OpenCV functions to compute gradient.

After your implementation, run the image_gradient.py in Python to verify it. Figure 1 shows an example of my implementation.

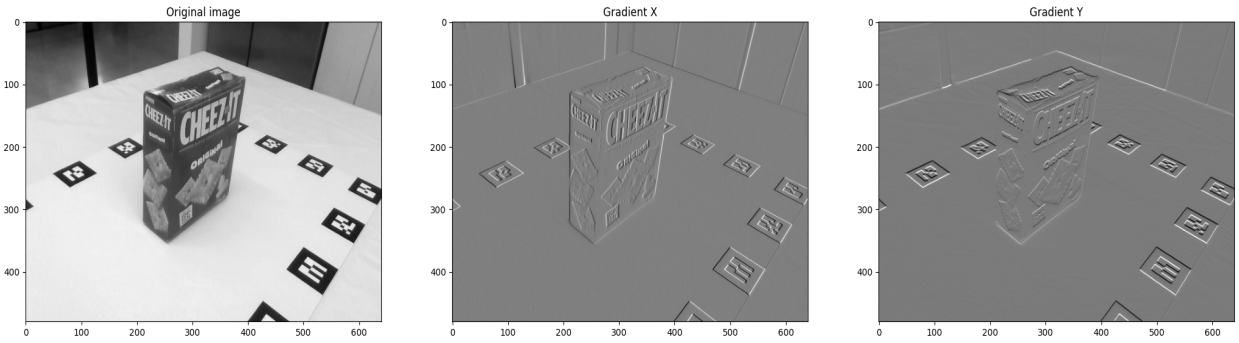



Figure 1: Image gradient.

# Problem 2

(5 points) Image filtering.

Implement the main() function in image_filtering.py.

This script first reads the "cracker_box.jpg" image, and then performs image filtering using the following $3 \times 3$ kernel:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \tag{2.1}$$

Note: in this implementation, you cannot directly use OpenCV functions to do the filtering.

After your implementation, run the image_filtering.py in Python to verify it. Figure 2 shows an example of my implementation.

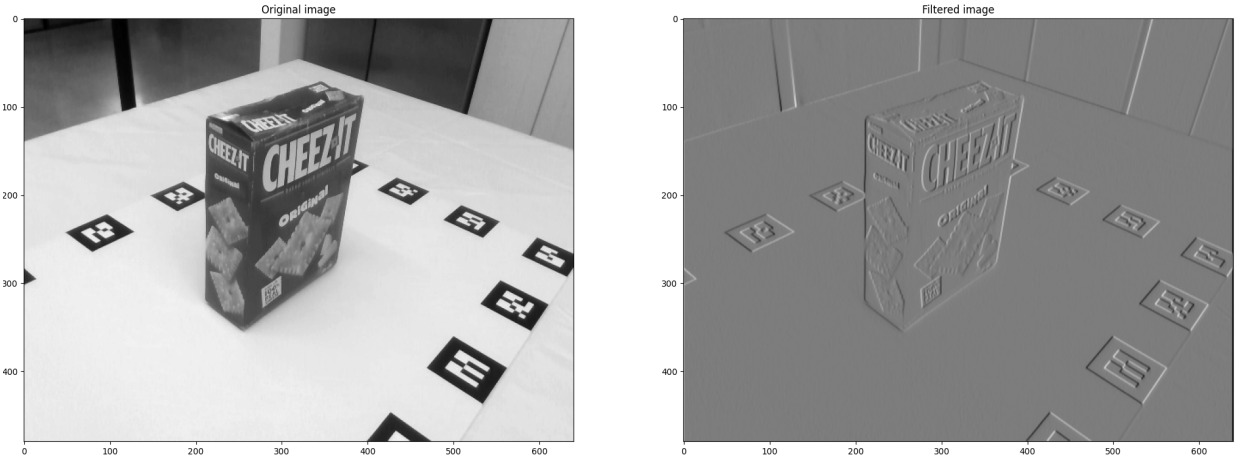



Figure 2: Image filtering.